

Homework 5

40 points

1. Find the rank of the following matrices **2*5 = 10 points**

$$\begin{bmatrix} 0 & 2 & 1 \\ 1 & 4 & 3 \\ 2 & 8 & 7 \\ 1 & 1 & 1 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$$

①

$$\begin{bmatrix} 0 & 2 & 1 \\ 1 & 4 & 3 \\ 2 & 8 & 7 \end{bmatrix} \xrightarrow{R_2: R_2 - 2R_1} \begin{bmatrix} 0 & 2 & 1 \\ 1 & 0 & 1 \\ 2 & 8 & 7 \end{bmatrix} \xrightarrow{R_3: R_3 - 4R_1} \begin{bmatrix} 0 & 2 & 1 \\ 1 & 0 & 1 \\ 2 & 0 & 3 \end{bmatrix} \xrightarrow{R_3: R_3 - 2R_2} \begin{bmatrix} 0 & 2 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$
$$\xrightarrow{R_2: R_2 - R_3} \begin{bmatrix} 0 & 2 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{swap } R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$\therefore \text{Rank} = 3$

②

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix} \xrightarrow{R_2 = R_2 - 2R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 3 & 5 & 7 \end{bmatrix} \xrightarrow{R_3 = R_3 - 3R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 2 & 4 \end{bmatrix} \xrightarrow{R_3 = R_3 - 2R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore \text{Rank} = 2$

2. What is the maximum possible rank of a 4×6 matrix? **2 points**

Maximum possible rank = $\min(\text{number of rows, number of columns})$

$$\therefore \min(4, 6) = 4$$

3. For what value of k does the following matrix have rank 2? **3 points**

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 2 & k \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 2 & k \end{bmatrix} \xrightarrow{R_2 = R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 3 & 2 & k \end{bmatrix} \xrightarrow{R_3 = R_3 - 3R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -4 & k-9 \end{bmatrix} \xrightarrow{\text{swap } R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -4 & k-9 \\ 0 & 0 & 0 \end{bmatrix}$$

No matter what value k takes in, the matrix will have a rank of 2

4. If A is a 2×3 matrix, what is the order (dimensions) of A^T ? **2 points**

$$A : 2 \times 3$$

$$A^T : 3 \times 2$$

5. Is the following matrix an orthogonal matrix? Show your steps **3 points**

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ -1 & 1 & 2 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ -1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 1 & -2 \\ 1 & 14 & 5 \\ -2 & 5 & 5 \end{bmatrix}$$

$$\therefore AA^T \neq I$$

$\therefore A$ is not an orthogonal matrix

6. What is the determinant of a square matrix with 2 identical rows? **2 points**

Square matrix A : $\begin{bmatrix} x & y \\ x & y \end{bmatrix} \xrightarrow{R_2: R_2 - R_1} \begin{bmatrix} x & y \\ 0 & 0 \end{bmatrix}$

$$\det(A) = 0 - 0 = 0$$

7. For what value of k will the determinant of the given matrix be 5? **3 points**

$$\begin{bmatrix} k & 1 \\ 3 & 2 \end{bmatrix}$$

$$\det \left(\begin{bmatrix} k & 1 \\ 3 & 2 \end{bmatrix} \right) = 2k - 3$$

$$2k - 3 = 5$$

$$2k = 8$$

$$k = 4$$

8. Find the determinant of: **5 points**

$$\begin{bmatrix} \overset{+}{1} & \overset{-}{2} & \overset{+}{0} & \overset{-}{0} \\ 0 & 3 & 4 & 0 \\ 0 & 0 & 5 & 6 \\ 0 & 0 & 0 & 7 \end{bmatrix}$$

Use co-factor expansion and mention the row/column used

$$\det = 1 \cdot \det \left(\begin{bmatrix} \overset{+}{3} & \overset{-}{4} & \overset{+}{0} \\ 0 & 5 & 6 \\ 0 & 0 & 7 \end{bmatrix} \right) - 2 \cdot \det \left(\begin{bmatrix} \overset{+}{0} & \overset{-}{4} & \overset{+}{0} \\ 0 & 5 & 6 \\ 0 & 0 & 7 \end{bmatrix} \right)$$

$$= 3 \begin{bmatrix} 5 & 6 \\ 0 & 7 \end{bmatrix} - 4 \begin{bmatrix} 0 & 6 \\ 0 & 7 \end{bmatrix} - 2 \left(-4 \begin{bmatrix} 0 & 6 \\ 0 & 7 \end{bmatrix} \right)$$

$$= 3 \cdot 35 - 0 - 0$$

$$= 105$$

9. Find the inverse of the following matrix: **6 points**

$$\begin{bmatrix} 3 & 0 & 2 \\ 2 & 0 & -2 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\begin{aligned} & \left[\begin{array}{ccc|ccc} 3 & 0 & 2 & 1 & 0 & 0 \\ 2 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_1 = R_1 + R_2} \left[\begin{array}{ccc|ccc} 5 & 0 & 0 & 1 & 1 & 0 \\ 2 & 0 & -2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \\ & \xrightarrow{\substack{R_1 = R_1/5 \\ R_2 = R_2/2}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{5} & \frac{1}{5} & 0 \\ 1 & 0 & -1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_2 = (R_2 - R_1) \times 10} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{5} & \frac{1}{5} & 0 \\ 0 & 0 & 1 & \frac{1}{5} & -\frac{3}{10} & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \\ & \xrightarrow{R_3 = R_3 - R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{5} & \frac{1}{5} & 0 \\ 0 & 0 & 1 & \frac{1}{5} & -\frac{3}{10} & 0 \\ 0 & 1 & 0 & -\frac{1}{5} & \frac{3}{10} & 1 \end{array} \right] \xrightarrow{\text{swap } R_3 \leftrightarrow R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{5} & \frac{1}{5} & 0 \\ 0 & 1 & 0 & -\frac{1}{5} & \frac{3}{10} & 1 \\ 0 & 0 & 1 & \frac{1}{5} & -\frac{3}{10} & 0 \end{array} \right] \end{aligned}$$

$\therefore$ Inversed matrix is $\begin{bmatrix} \frac{1}{5} & \frac{1}{5} & 0 \\ -\frac{1}{5} & \frac{3}{10} & 1 \\ \frac{1}{5} & -\frac{3}{10} & 0 \end{bmatrix}$

* Determinant method is also valid to use

10. For what value of k is the given matrix invertible? **4 points**

$$\begin{bmatrix} 1 & 2 \\ 3 & k \end{bmatrix}$$

Explain your answer

In order to be invertible, determinant has to be non-zero

$$\therefore k - 6 \neq 0$$

$$k \neq 6$$